New Hampshire 2026 SGP Results

NH SAS (Grades 3–8) and WIDA ACCESS (Grades K–12)

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NH TAC Meeting · October 19, 2026

AI Contribution & Authorship Statement

These analyses and this presentation represent the work of the author with substantive and continuous assistance from AI-based systems (Anthropic's Claude), integrated natively into the analysis, revision, and preparation of this deck.

> **The use of AI here is not incidental or peripheral — it is essential, embedded, and reflective of a methodological shift toward AI-native scholarship. This statement is offered in the spirit of methodological clarity and epistemic honesty.**

- **Verification of inputs:** Every figure and factual claim is generated from cached aggregates of the student-level files. Claims that cannot be verified from those files are deliberately **not asserted**.
- **Computational modules:** AI contributed to the R summary pipeline, the `ggplot2` figures, and the verification of reported numbers against the analysis logs.
- **Authorship & responsibility:** The author retains full responsibility for all claims, analyses, and recommendations (ICMJE/AERA norms). AI is not listed as an author but its instrumental role is explicitly acknowledged.

Roadmap

1. **What was run** — NH SAS cohort and baseline analyses
2. **2026 results** — overall growth and model fit
3. **Growth in context** — pandemic recovery and the 2018–19 baseline
4. **Growth equity** — results by student group
5. **Target attainment** — catch-up / keep-up
6. **WIDA ACCESS** — growth to the Level 4.5 exit standard
7. **Questions for the TAC**

> **110,854** students received a 2026 NH SAS growth percentile (grades 4–8, ELA and mathematics). **4,404** English learners received a 2026 WIDA baseline growth percentile. NH SAS results use the student-level file received **July 9, 2026**.

SGP results for 2026

What was run: NH SAS (grades 4–8)

Two SGP analyses were conducted on the 2026 NH SAS data (ELA and Mathematics, grades 3–8; growth from grade 4):

- **Cohort-referenced** — growth norms built from the most recent cohorts; by construction these center mean/median SGP on **50**.
- **Baseline-referenced** — historical coefficient matrices from the pre-pandemic **2018–19** administration (with 2017–18 priors). These are **not** centered on 50 and let us examine COVID-era academic recovery, changes in rates of learning, and possible **scale drift**.
- **Projections** (lagged and straight, three-year catch-up / keep-up and move-up / stay-up) were calculated using both norm sets.

Science and grade 11 are on the file but are **not** included in the SGP analyses.

> Cohort norms answer *"how did this student grow relative to academic peers this year?"* Baseline norms answer *"how does that growth compare to pre-pandemic rates of learning?"*

Provenance & reproducibility

- NH SAS analyses used student-level data supplied by the New Hampshire Department of Education, **received July 9, 2026**.
- WIDA ACCESS analyses used the ESOL extract for 2026 (with both the new and old composite scales).
- Conducted with the **SGP R package** (version 2.3.1.16).
- All analyses converged; goodness-of-fit plots confirm successful analyses.
- Source code is public on GitHub:
  - NH SAS: github.com/CenterForAssessment/New_Hampshire
  - WIDA ACCESS: github.com/CenterForAssessment/WIDA_NH
  - SGP package: github.com/CenterForAssessment/SGP

Reproducible pipeline

This deck is generated from cached summaries of the student-level LONG data. When NHDOE supplies revised data, re-running the pipeline refreshes every number and table automatically.

2026 SGP results at a glance

Grade	ELA		Mathematics	
	Cohort	Baseline	Cohort	Baseline
4	49.9/50/11,153	52.2/53/11,153	49.9/50/11,225	51.3/51/11,225
5	50.0/50/11,092	54.7/56/11,092	50.0/50/11,148	49.3/49/11,148
6	49.8/50/11,045	51.7/52/11,045	49.8/50/11,122	51.4/52/11,122
7	50.0/50/11,106	57.2/61/11,106	49.9/50/11,210	54.1/56/11,210
8	49.9/50/10,825	54.2/56/10,825	49.8/50/10,928	54.9/57/10,928

Each cell: mean/median/N of student growth percentiles, 2026. Cohort norms center on 50 by construction; baseline norms are referenced to 2018–19 rates of learning. Data received July 9, 2026.

Reading the 2026 baseline results

Above pre-pandemic rates (baseline mean > 52)

- ELA Grade 7 (mean **57.2**)
- Mathematics Grade 8 (mean **54.9**)
- ELA Grade 5 (mean **54.7**)
- ELA Grade 8 (mean **54.2**)
- Mathematics Grade 7 (mean **54.1**)
- ELA Grade 4 (mean **52.2**)

Below pre-pandemic rates (baseline mean < 48)

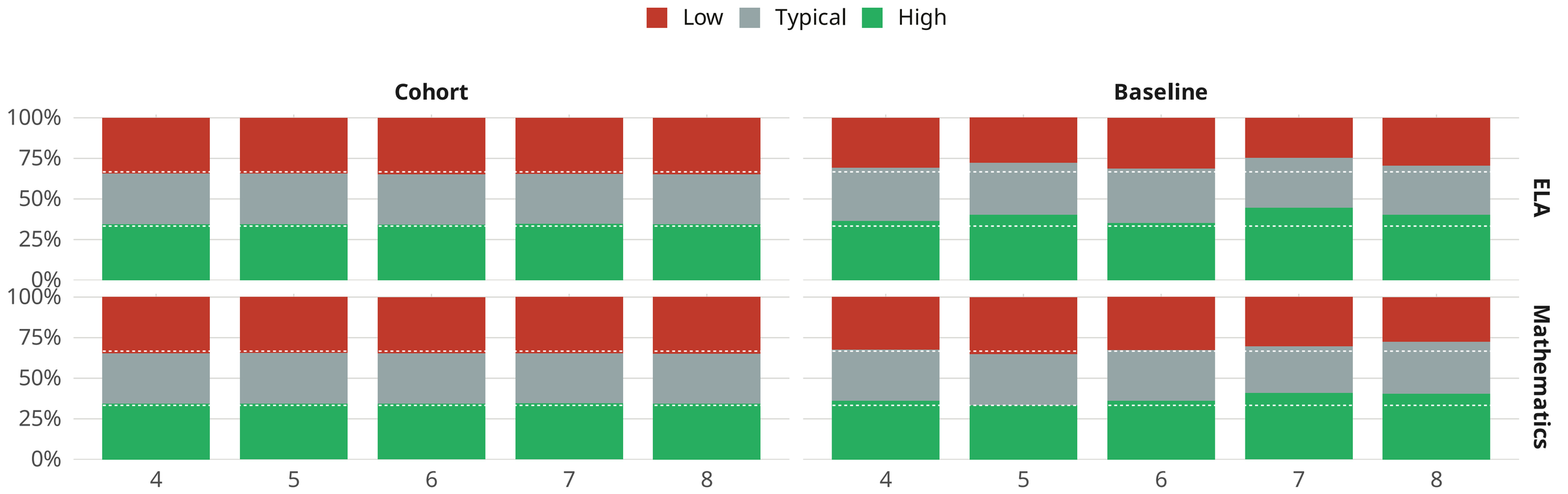
- None materially different

Cohort-referenced results are, as expected, centered on 50 with even conditional coverage. The **baseline** results are the diagnostic lens — they say whether 2026 growth matches the 2018–19 rate of learning.

Growth distribution: Low / Typical / High

Share of students at each growth level, 2026

Cohort norms split roughly one-third each; baseline shifts reveal recovery / drift



Model fit and conditional coverage

What these plots do and do not diagnose

Each plot shows, for every decile of prior achievement, the share of students falling in each growth-percentile decile. Two different things move those cells, and only one of them is a fitting problem.

What they diagnose

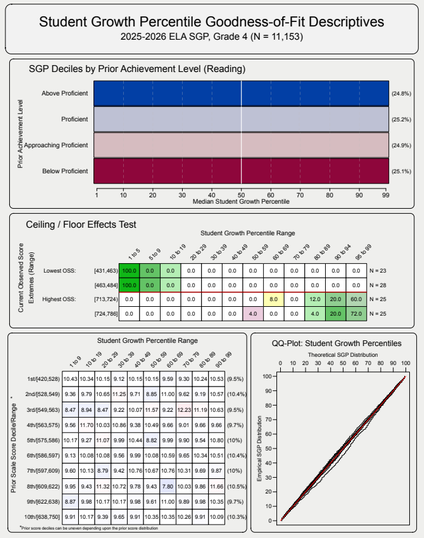
Systematic tilt across prior achievement. If growth percentiles drift upward or downward as prior achievement rises, the conditional model is not covering the range of prior achievement evenly. That is misfit, and it is actionable.

What they cannot diagnose

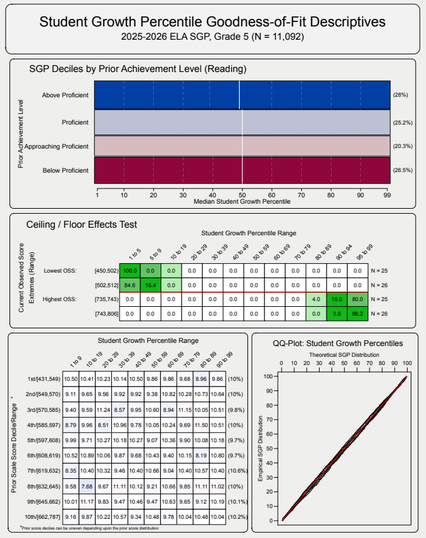
Departure from exact uniformity. Scores are discrete. Students sharing a prior and a current score must receive the same growth percentile, so the conditional distribution cannot be uniform and the cells cannot all sit at 10%. That lumpiness follows from fitting a continuous quantile model to a discrete scale.

> The diagnostic content is the **tilt**, not the distance from 10%. For baseline norms a level shift is neither misfit nor uniformity: it is the change in the prior-to-current relationship that the comparison exists to detect.

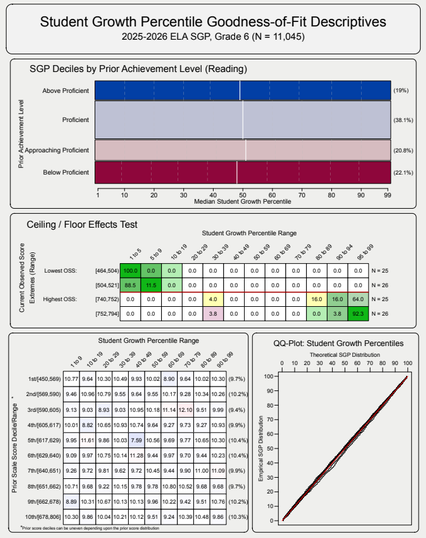
NH SAS ELA – cohort model fit, 2026



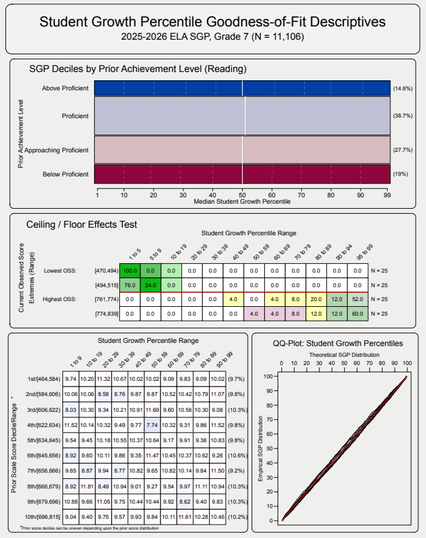
ELA Grade 4



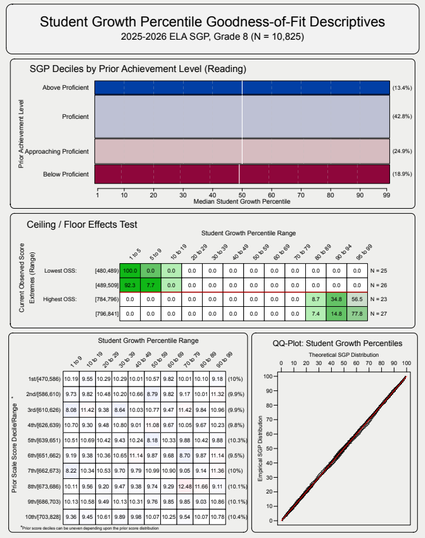
ELA Grade 5



ELA Grade 6

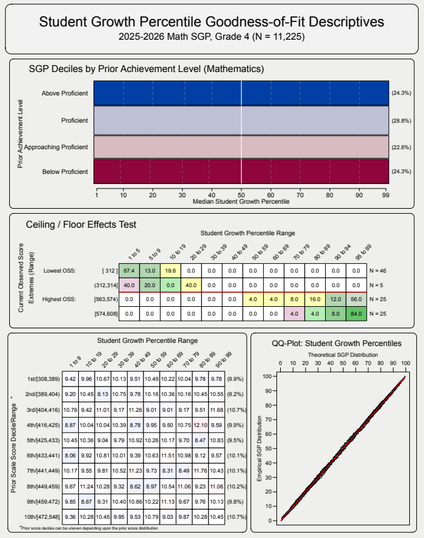


ELA Grade 7

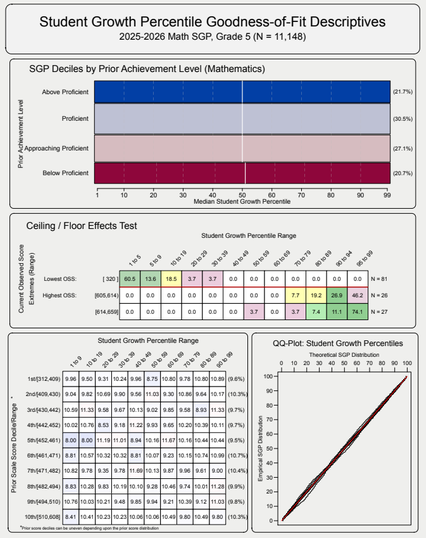


ELA Grade 8

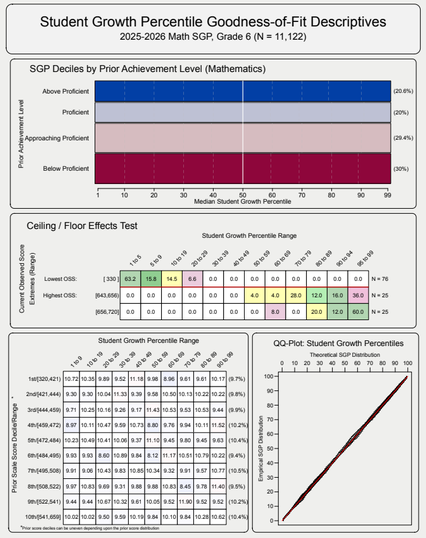
NH SAS Mathematics – cohort model fit, 2026



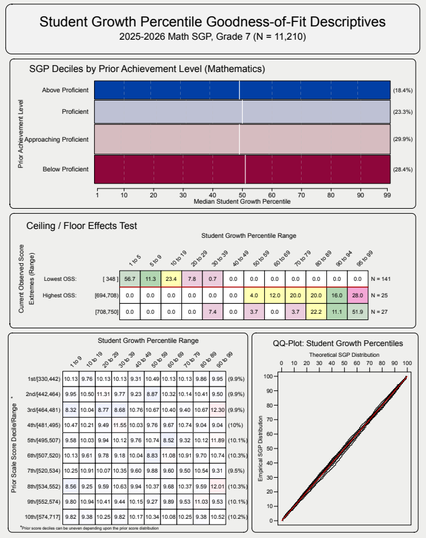
Mathematics Grade 4



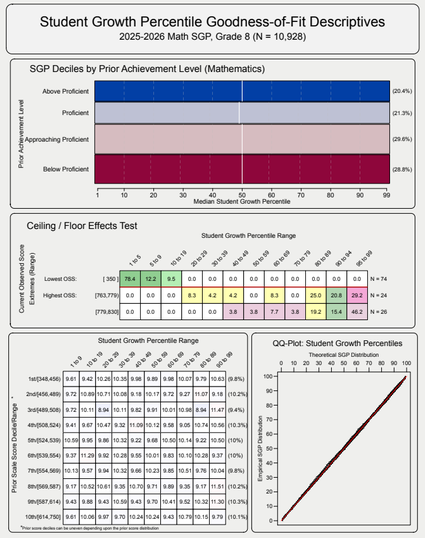
Mathematics Grade 5



Mathematics Grade 6

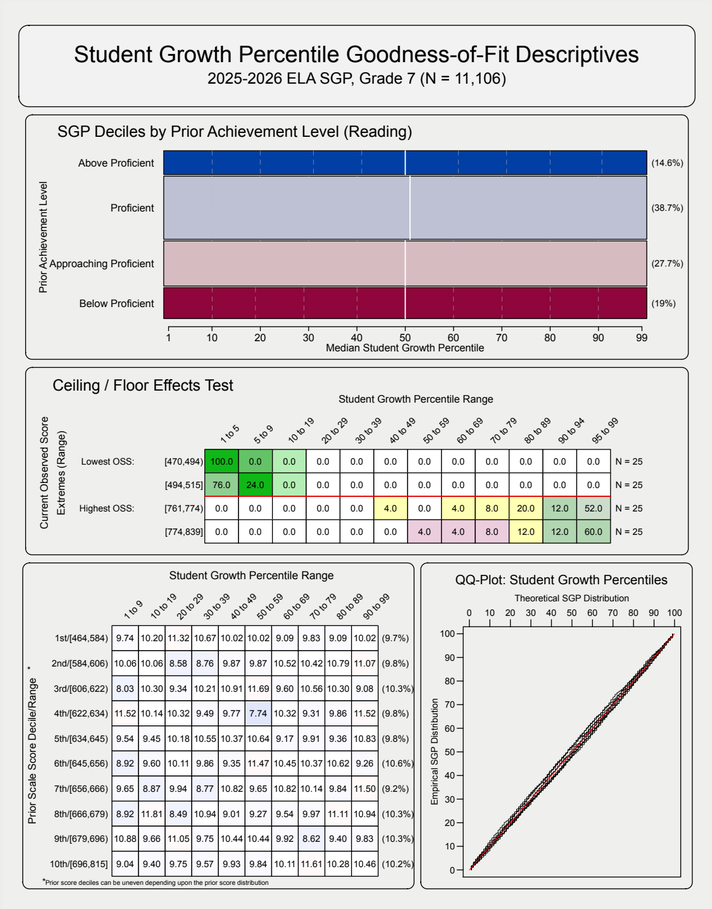


Mathematics Grade 7

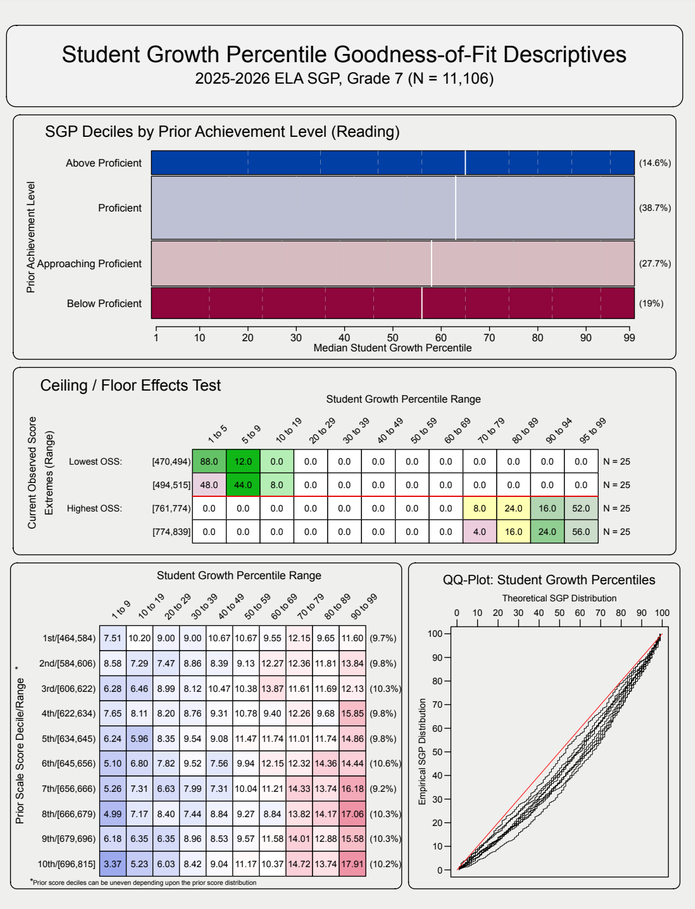


Mathematics Grade 8

Cohort vs. baseline fit — ELA Grade 7



ELA Grade 7 — Cohort norms



ELA Grade 7 — Baseline norms

Growth in context

Where were they during the pandemic?

The 2026 tested grades sat in very different places when schooling was disrupted. There is no `2019_2020` NH SAS administration.

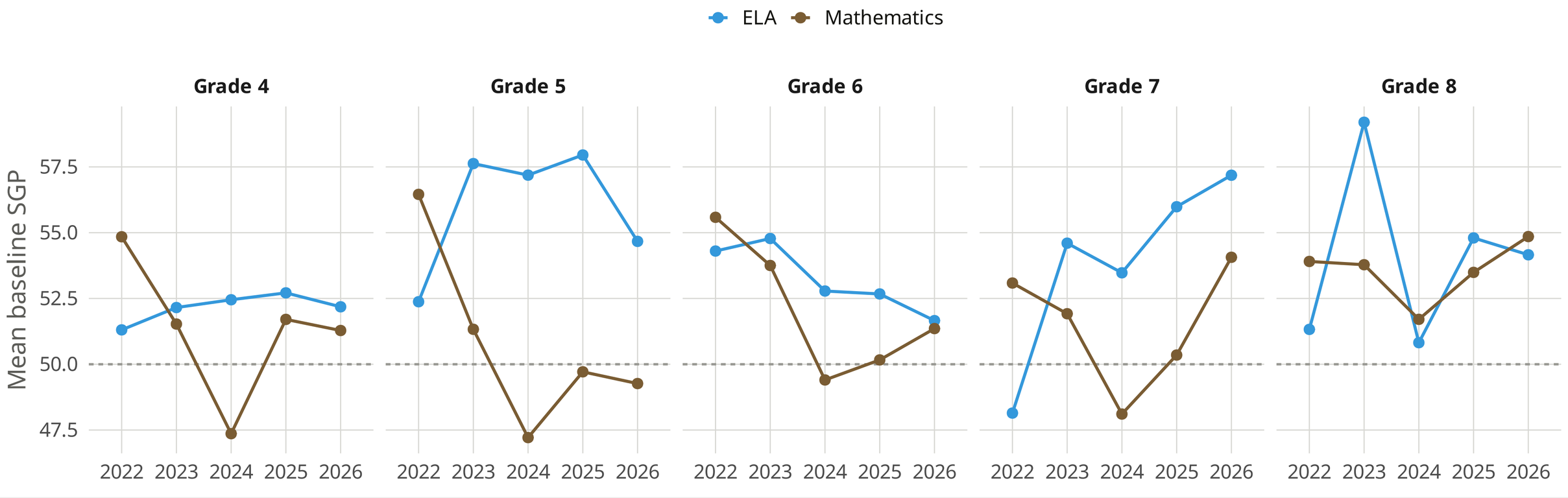
Grade in 2026	Spring 2020	2020–21 (first post-COVID test)
Grade 8	Grade 2	Grade 3
Grade 7	Grade 1	Grade 2
Grade 6	Kindergarten	Grade 1
Grade 5	Pre-K	Kindergarten
Grade 4	Not yet in school	Pre-K

Grade 8 in 2026 spent early-elementary years in disrupted schooling; grade 4 had not yet entered the tested progression.

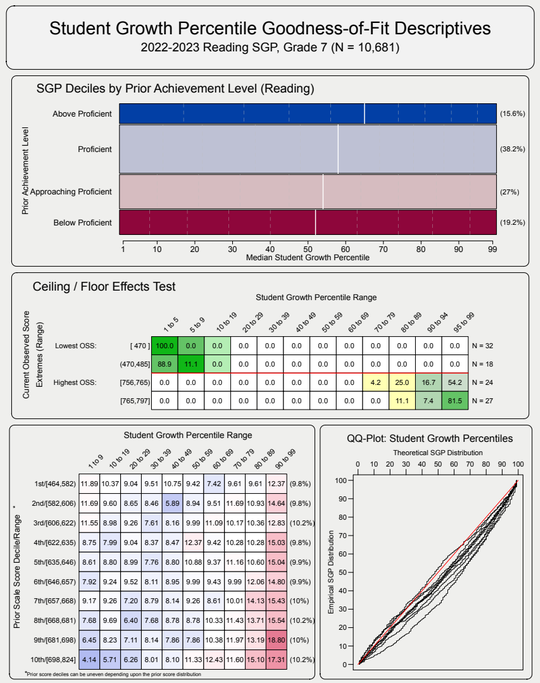
Trend: same grade over time (baseline SGP)

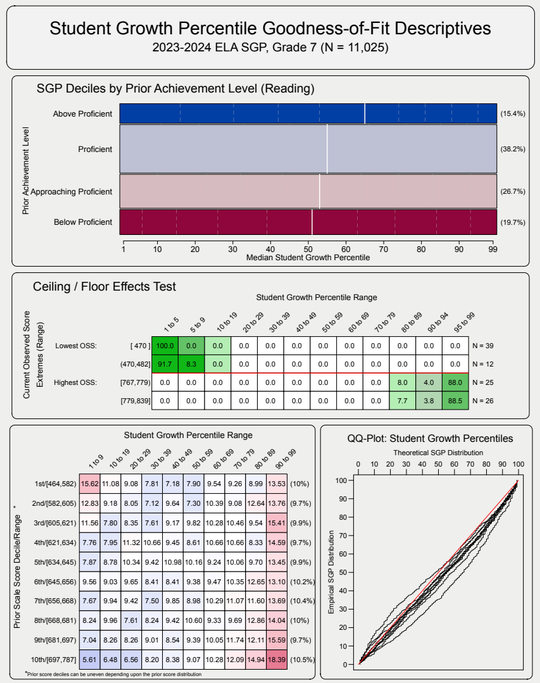
Same-grade baseline-referenced growth, 2022–2026

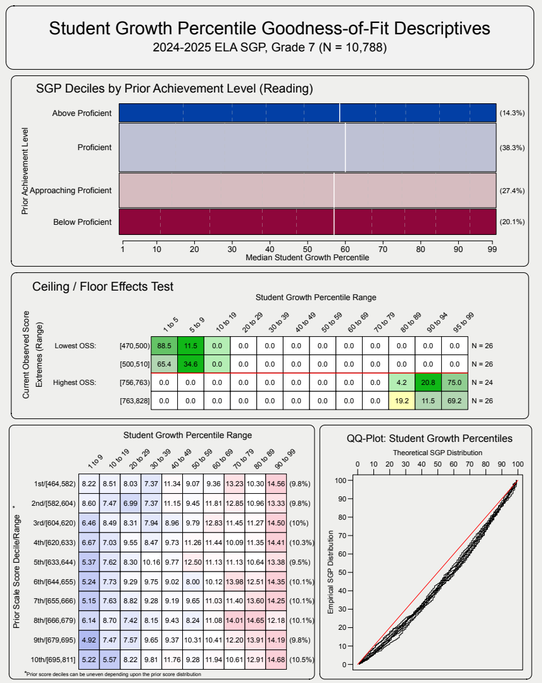
Dashed line = the 2018–19 rate of learning. A return toward 50 is recovery completing, not a new decline.

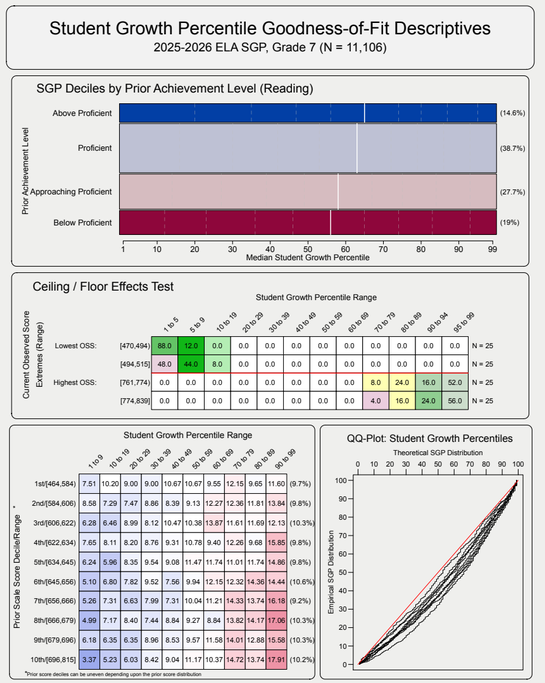


ELA Grade 7 over time – goodness of fit









ELA Grade 7 — 2023

ELA Grade 7 — 2024

ELA Grade 7 — 2025

ELA Grade 7 — 2026

Growth equity

Growth by student group (NH SAS, 2026)

Group	ELA		Mathematics	
	Cohort	Baseline	Cohort	Baseline
English Learner				
Non-English Learners (Non-EL)	50.0/50/28.9/52,303	54.1/56/28.3/52,303	49.9/50/28.9/52,519	52.2/53/29.0/52,519
English Learners (EL)	49.1/48/28.6/2,918	51.8/53/28.1/2,918	49.5/50/28.9/3,114	51.1/51/28.8/3,114
Students w/ Disability				
Students without Disabilities (Non-SWD)	51.4/52/28.7/44,185	55.9/58/28.0/44,185	50.9/51/28.7/44,487	53.4/55/28.8/44,487
Students with Disabilities (SWD)	44.1/42/28.7/11,036	46.4/45/28.2/11,036	45.8/44/29.3/11,146	47.2/46/29.2/11,146
Economically Disadv.				
Economically Disadvantaged	46.9/46/29.0/13,673	50.0/50/28.5/13,673	46.4/45/29.0/13,886	48.1/47/29.1/13,886
Not Economically Disadvantaged	50.9/51/28.8/41,548	55.3/57/28.1/41,548	51.0/52/28.8/41,747	53.5/55/28.8/41,747

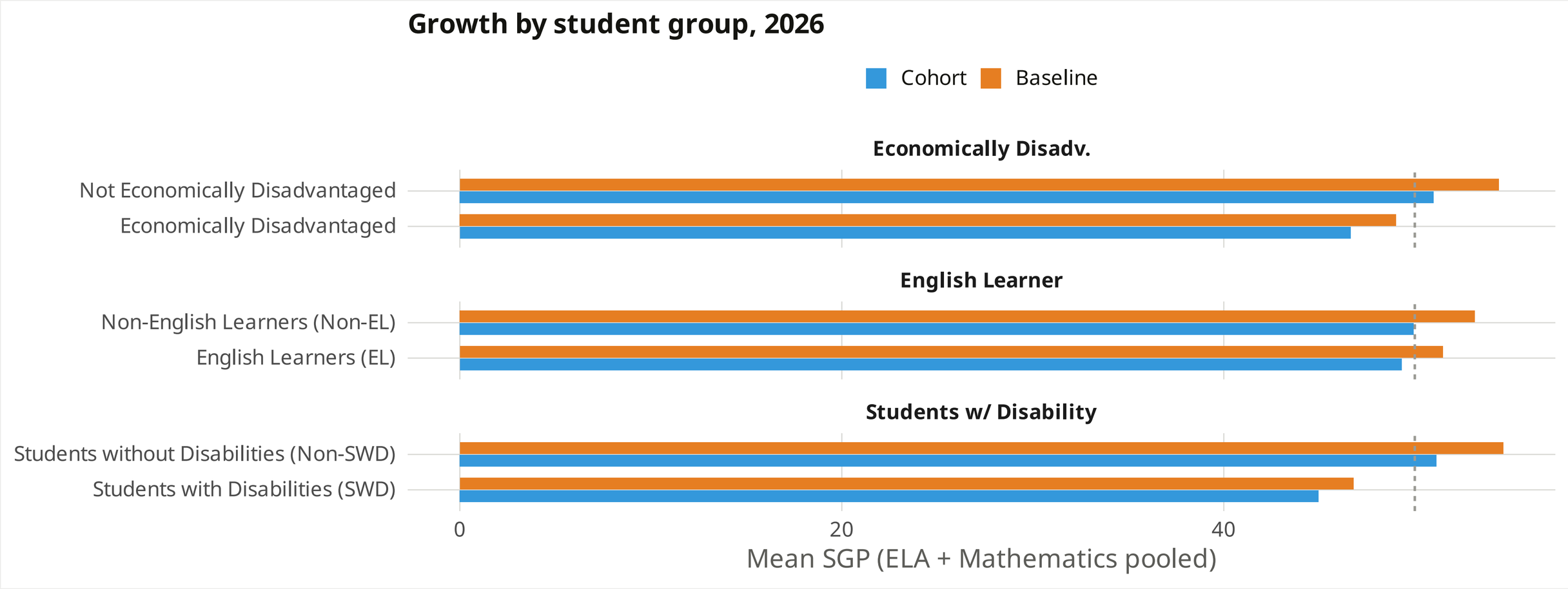
mean/median/SD/N, NH SAS grades 4–8 pooled, 2026. Cells with N < 10 are suppressed. For reference, SGPs uniformly distributed over the integers 1–99 have SD = $\sqrt{((99^2 - 1)/12)} \approx 28.6$.

Growth by gender and race/ethnicity (NH SAS, 2026)

Group	ELA		Mathematics	
	Cohort	Baseline	Cohort	Baseline
Gender				
Male	48.5/48/29.2/28,330	52.3/53/28.7/28,330	49.9/50/29.2/28,562	52.2/53/29.3/28,562
Female	51.4/52/28.5/26,815	55.7/57/27.7/26,815	49.9/50/28.5/26,994	52.1/53/28.6/26,994
Race/Ethnicity				
White	49.9/50/28.8/45,358	54.0/55/28.2/45,358	50.0/50/28.9/45,567	52.3/53/29.0/45,567
Asian	55.3/57/28.6/1,566	59.8/63/27.6/1,566	56.0/58/28.2/1,610	58.9/62/28.0/1,610
Black or African American	48.0/48/29.1/1,080	51.0/52/28.7/1,080	47.3/47/28.0/1,111	48.7/49/28.0/1,111
Hispanic or Latino	48.6/48/29.2/4,824	51.7/53/28.7/4,824	47.2/46/29.0/4,943	49.0/48/29.0/4,943
Two or More Races	50.5/51/28.6/2,191	54.9/57/27.8/2,191	50.2/51/29.0/2,199	52.6/53/29.1/2,199
American Indian or Alaska Native	51.2/52/28.8/94	55.1/56/28.4/94	55.1/58/29.5/93	57.5/62/29.3/93
Native Hawaiian or Pacific Islander	49.2/48/26.6/32	54.2/56/25.7/32	54.2/58/26.0/33	57.4/63/24.8/33

mean/median/SD/N, NH SAS grades 4–8 pooled, 2026. Native Hawaiian or Pacific Islander and American Indian or Alaska Native have small N; cells with N < 10 are suppressed. For reference, SGPs uniformly distributed over the integers 1–99 have SD = $\sqrt{((99^2 - 1)/12)} \approx 28.6$.

Equity gaps at a glance



Within-year growth is close to comparable across groups

Group vs. comparison group	ELA	Mathematics
Gender (Female vs Male)	51.4 vs 48.5	49.9 vs 49.9
English learners	49.1 vs 50.0	49.5 vs 49.9
Students with disabilities	44.1 vs 51.4	45.8 vs 50.9
Economically disadvantaged	46.9 vs 50.9	46.4 vs 51.0

Mean cohort-referenced SGP, NH SAS grades 4–8 pooled, 2026.

Where achievement gaps are large, **within-year growth is often not**. Cohort SGPs near 50 for both members of a pair mean the gap is being maintained rather than widened this year.

Target attainment

What the target actually is

“Catch up” and “keep up” mean nothing without a stated horizon. In New Hampshire the NH SAS target is proficiency **by the end of the grades 3–8 progression**, so the horizon shrinks as students approach grade 8.

Grade	Horizon to target	Catch-up rate	Keep-up rate
Grade 4	3 years	36.0%	65.4%
Grade 5	3 years	34.2%	63.7%
Grade 6	2 years	28.8%	68.1%
Grade 7	1 year	23.3%	76.4%
Grade 8	same year	13.8%	80.7%

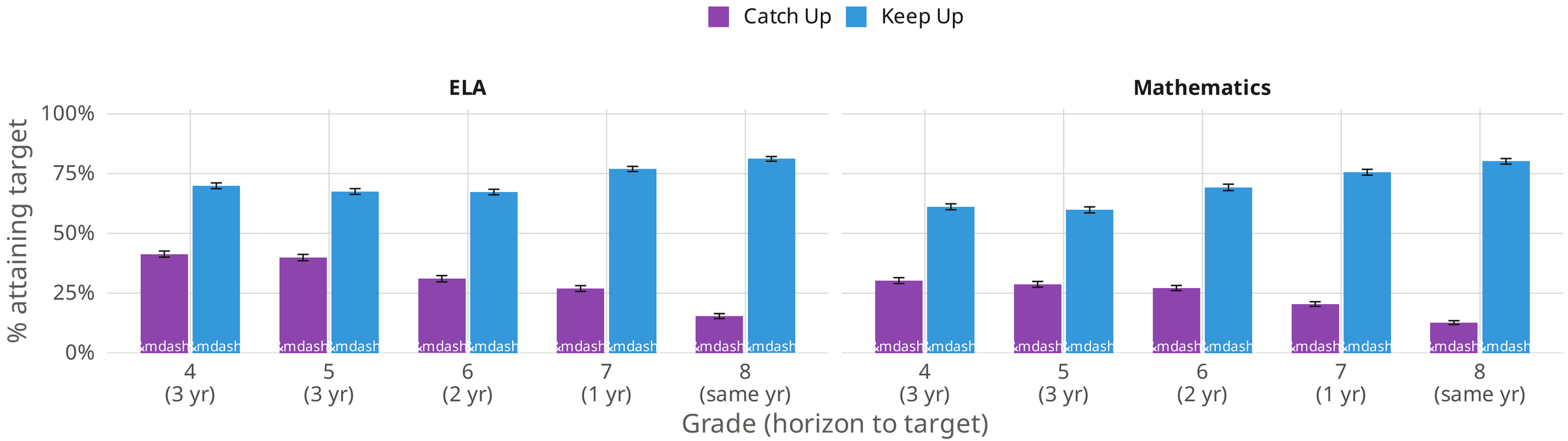
Cohort targets, NH SAS, ELA and mathematics pooled, 2026. Horizon is the median number of years to the target.

As the horizon shrinks, catch-up rates fall and keep-up rates rise. Most of the apparent grade gradient is the target, not the students.

Catch-up / keep-up attainment (NH SAS, 2026)

Share of students on track to catch up / keep up

Catch Up = students below proficient; Keep Up = students at/above.
Bars show 95% Wilson intervals; N is printed in each bar.



Target attainment by prior achievement

16%

of students who were **Below Proficient** last year are on track to **catch up**

N = 27,027

82%

of students who were **Above Proficient** last year are on track to **keep up**

N = 22,775

These are different students judged against different targets, so the two figures are not a gap. NH SAS, ELA and mathematics pooled, 2026.

Students entering below proficient face the steepest catch-up targets and attain them least often — the central equity concern for accountability use of these growth results.

WIDA ACCESS for New Hampshire

What was run: WIDA ACCESS

English learners in New Hampshire take ACCESS for ELLs. The SGP analyses treat the Overall ELP composite as `READING` and ask whether students are **on track to the state's exit standard**.

- **Exit / proficiency criterion: WIDA Level 4.5** (in `SGPstateData` and in the prep scripts since 2019). Fine-grained scores 4.5–4.9 are collapsed to Level 4.5.
- **Cohort SGPs** for 2026 were computed on the **new ACCESS 3.0 scale**.
- **Baseline SGPs and baseline catch-up / keep-up targets** were computed on the **old ACCESS 2.0 scale**, against the existing 4.5 cuts. Those are the headline on-track results in this section.
- Targets run 1 to 6 years forward. The rates below use the **6-year** horizon — the longest window present in the object.

2026 is a transition year

`WIDA_NH` does not yet carry the consortium's 2026 transition metadata (new cuts, knots, linkages). Cohort targets computed on the new scale against old-scale cuts are not on the same footing as the baseline targets. They are shown only as a discussion item.

ACCESS participation by grade, 2026

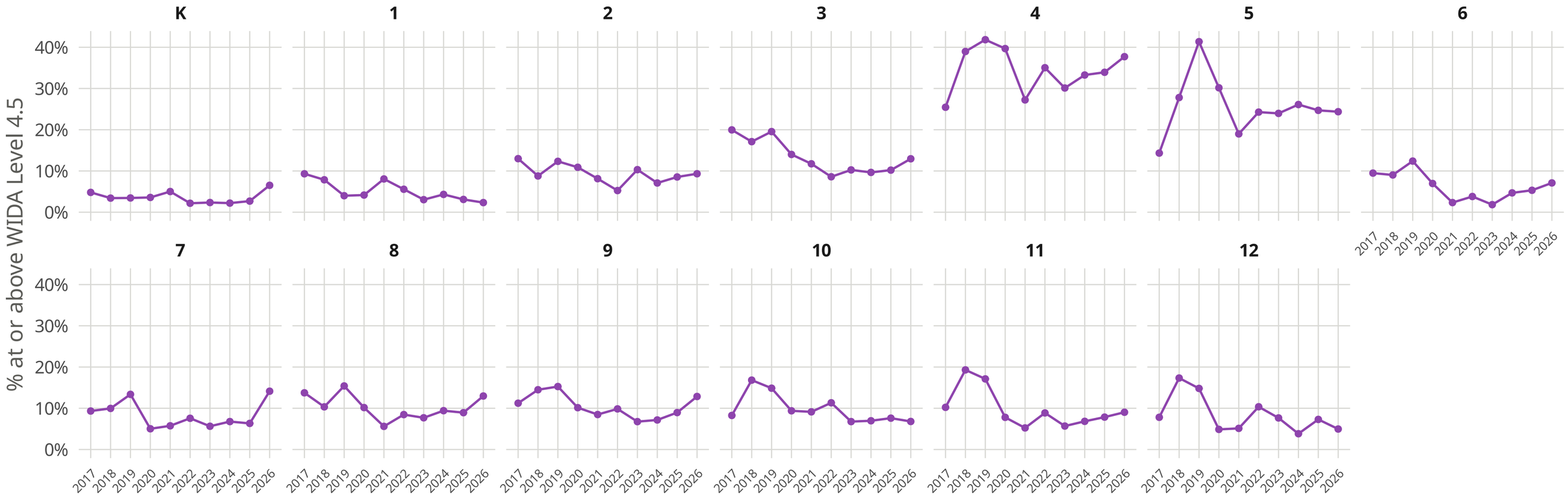
Grade	N tested	N with cohort SGP	N with baseline SGP
K	506	0	0
1	557	443	443
2	611	511	511
3	602	522	522
4	631	564	564
5	398	337	337
6	352	286	286
7	389	318	318
8	370	320	320
9	420	336	336
10	353	286	286
11	321	256	256
12	302	225	225

WIDA ACCESS Overall ELP, 2026. Kindergarten is shown as K. Cells with N < 10 are suppressed later in the on-track tables, not here.

Percent at or above Level 4.5 over time

Share of tested students at the New Hampshire exit standard

Reported achievement level. 2026 levels are the vendor-reported ACCESS 3.0 classifications.



2026 WIDA SGP at a glance

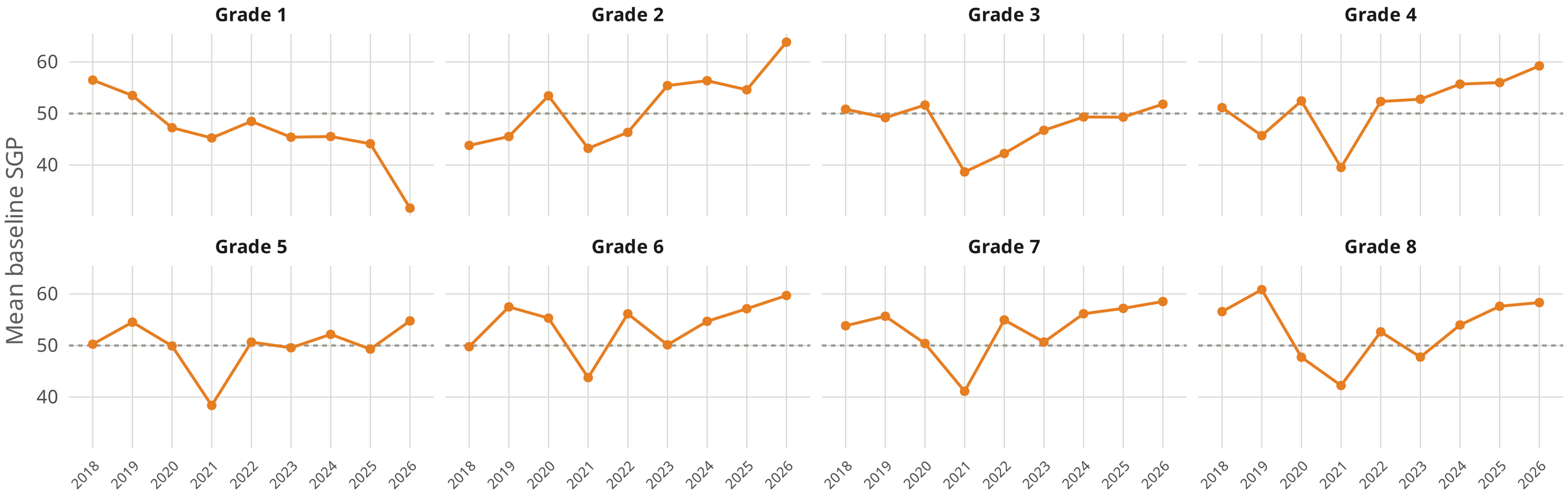
Grade	Cohort (new scale)	Baseline (old scale)
K	—	—
1	49.1/49/443	31.6/24/443
2	49.3/49/511	63.8/70/511
3	48.4/48/522	51.8/53/522
4	49.6/50/564	59.2/64/564
5	49.3/50/337	54.8/57/337
6	48.1/47/286	59.7/63/286
7	49.1/50/318	58.5/65/318
8	49.1/48/320	58.3/64/320
9	49.0/51/336	56.9/60/336
10	48.4/48/286	53.7/60/286
11	49.4/50/256	53.9/54/256
12	47.9/49/225	51.4/54/225

mean/median/N, WIDA ACCESS Overall ELP, 2026. Baseline cells are the headline growth results for this transition year. Cells with N < 10 are suppressed.

Baseline SGP trend, WIDA ACCESS

WIDA ACCESS baseline-referenced growth, grades 1–8

2026 points use the old composite scale. Dashed line = the historical ACCESS 2.0 rate of learning.



The 2026 scale transition

Grade	N	Mean (new − old)	Mean |old − linked(new)|	4.5 cut old → new	% ≥ 4.5 (matched cut)	% ≥ 4.5 old cut on new score
K	506	-10.8	0.00	309 → 283	6.5%	2.8%
1	557	+36.3	0.00	330 → 374	2.3%	13.6%
2	611	+25.0	0.00	344 → 378	9.3%	42.2%
3	602	+28.8	0.00	356 → 394	13.0%	51.5%
4	631	+13.5	0.00	366 → 381	37.7%	56.9%
5	398	+12.7	0.00	374 → 391	24.4%	43.7%
6	352	+34.3	0.00	382 → 430	7.1%	46.0%
7	389	+37.5	0.00	389 → 439	14.1%	52.7%
8	370	+39.3	0.00	395 → 448	13.0%	55.7%
9	420	+40.1	0.00	402 → 457	12.9%	55.0%
10	353	+39.7	0.00	408 → 465	6.8%	53.3%
11	321	+40.6	0.00	413 → 471	9.0%	49.8%
12	302	+39.0	0.13	418 → 477	5.0%	43.0%

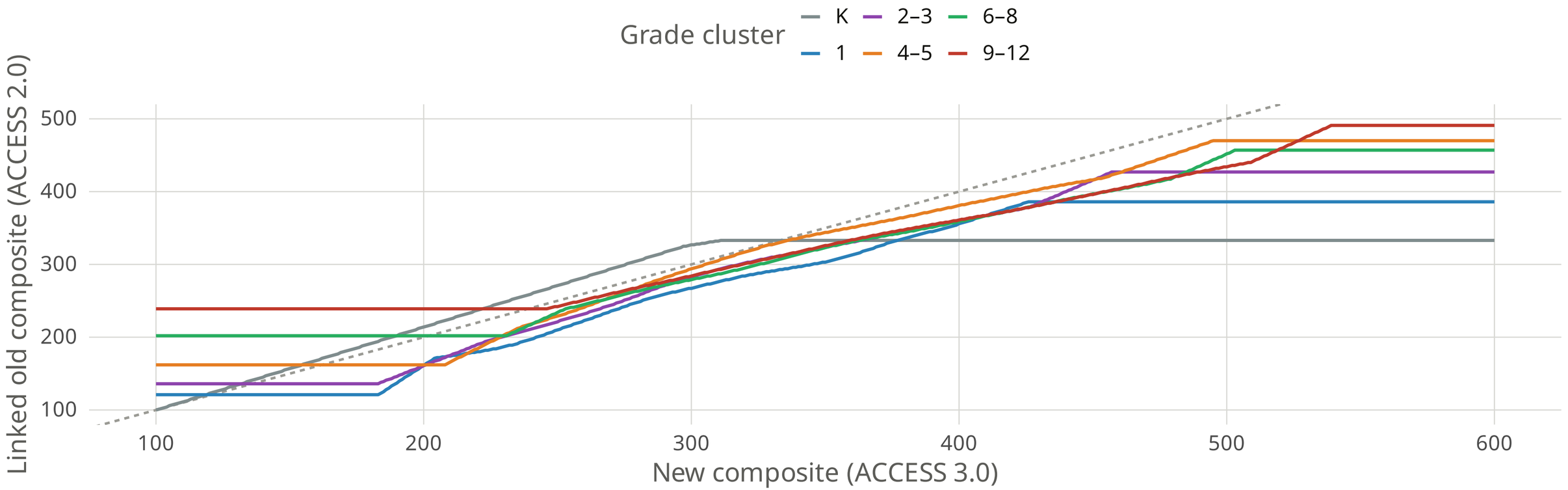
Both composites are scored 100–600; that is a shared range, not a shared scale. The vendor old-scale composite is the consortium NEW_to_OLD concordance of the new composite (largest grade mean gap 0.13 points), not a separate measurement. The new-scale 4.5 cut is the smallest new score that the concordance links to the old cut. The matched-cut rate is identical on either scale and agrees with the vendor-reported level for 5,812 of 5,812 tested students. The last column applies old cuts to new scores and is not a valid result. The concordance is the one on generic WIDA in SGPstateData; WIDA_NH does not yet carry it.

Levels carry across the transition; raw composites do not. WIDA's interim 2026 cuts are the concordance images of the old cuts, so a 2026 Level 4.5 means the same thing as a 2025 Level 4.5 to the extent the concordance holds. Mean (new − old) is not growth,

One concordance per ACCESS grade cluster

The consortium 2026 concordance has one table per grade cluster

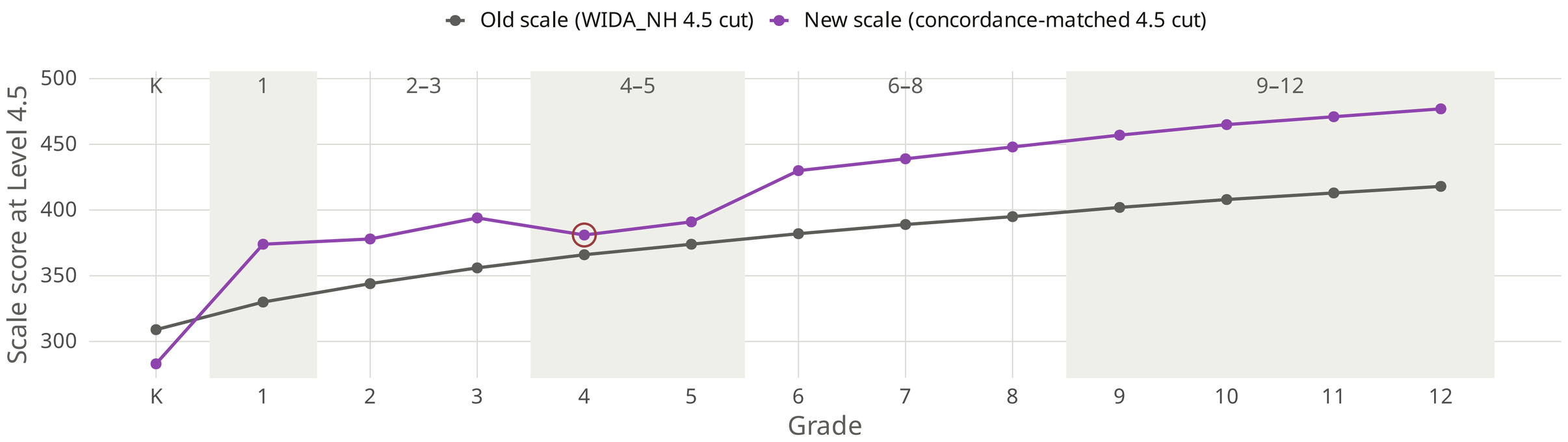
Each line gives the old score a new score links to. Dashed line = identity.
Flat runs are floors and ceilings, where many new scores share one old score.



The 4.5 exit cut on each scale

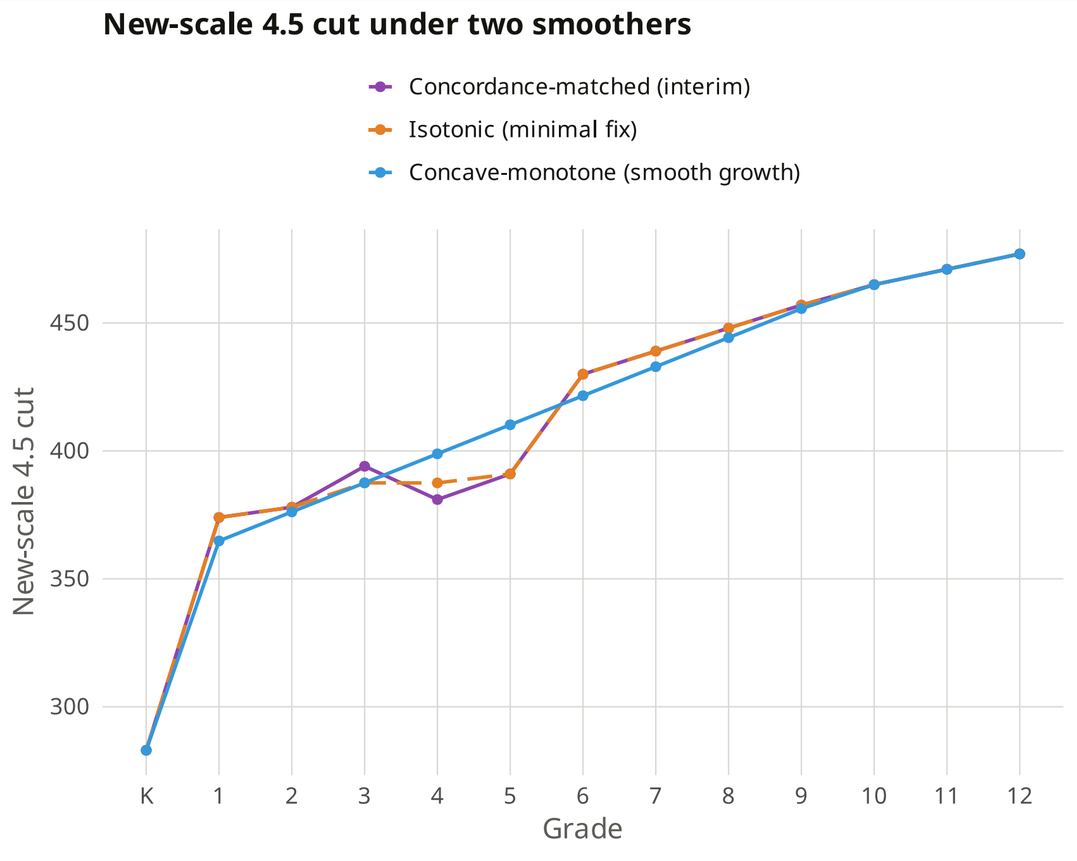
The old 4.5 cuts rise every grade; their new-scale images do not

Shaded bands mark the concordance grade clusters. Circled: a new cut below the grade before it.



The grade 3→4 dip is not new. On the old scale, grade 4 had the highest share at or above 4.5 in 9 of 9 years from 2017 to 2025 (25–42%, against 9–20% in grade 3). On the new scale, that long-standing seam between the grade 2–3 and 4–5 concordance tables shows up as a cut that falls.

What-if: standard setting smooths the new cuts

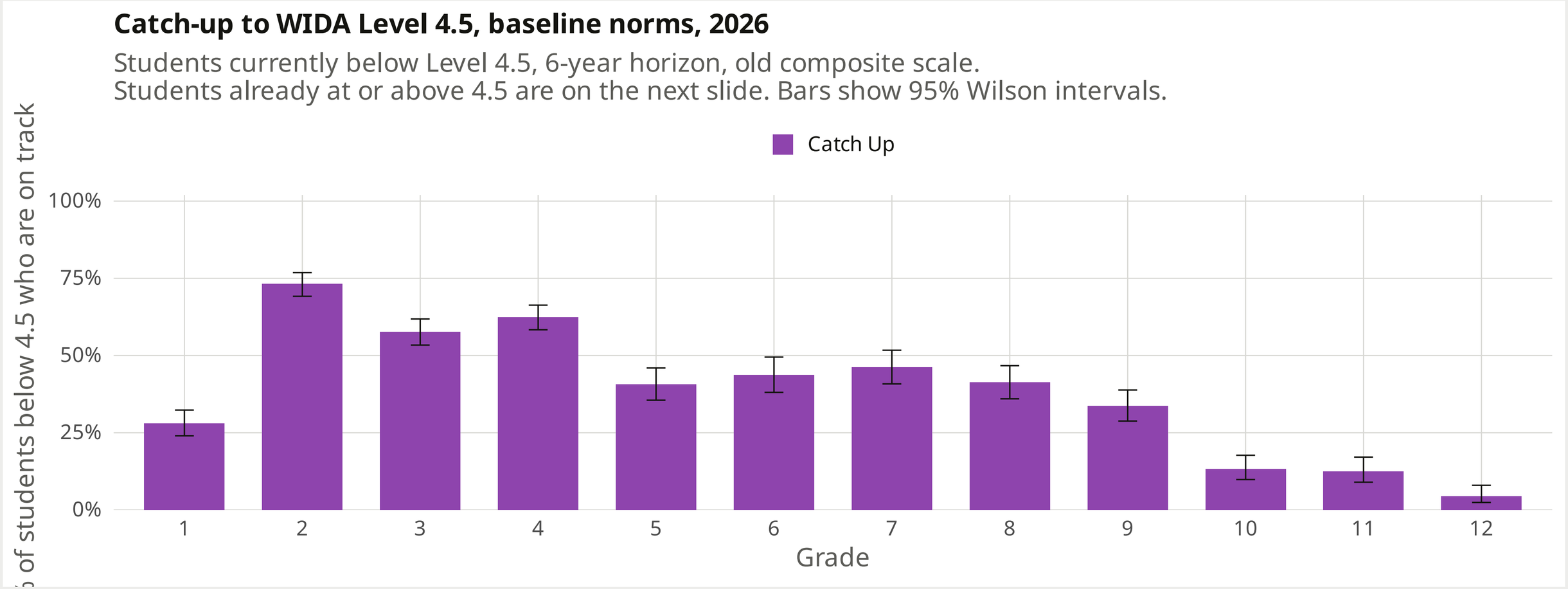


Grade	N	Interim	Isotonic	Concave
1	557	2.3%	2.3%	4.1%
2	611	9.3%	9.3%	10.1%
3	602	13.0%	17.9%	17.9%
4	631	37.7%	31.7%	21.7%
5	398	24.4%	24.4%	10.6%
6	352	7.1%	7.1%	10.2%
7	389	14.1%	14.1%	16.5%
8	370	13.0%	13.0%	14.6%
9	420	12.9%	12.9%	13.8%

% of 2026 students at or above 4.5 on the new composite under each set of cuts. Red cells move. Unchanged under both smoothers: grades K, 10, 11, 12. A what-if, not a prediction of WIDA's cuts.

Baseline growth and on-track results do not move: they use the old scale and the old 4.5 cuts. What moves is a 2026 exit decision made on the new composite, mostly in grades 3–5.

On track to Level 4.5 by grade (baseline, 6-year)



On track by current WIDA level

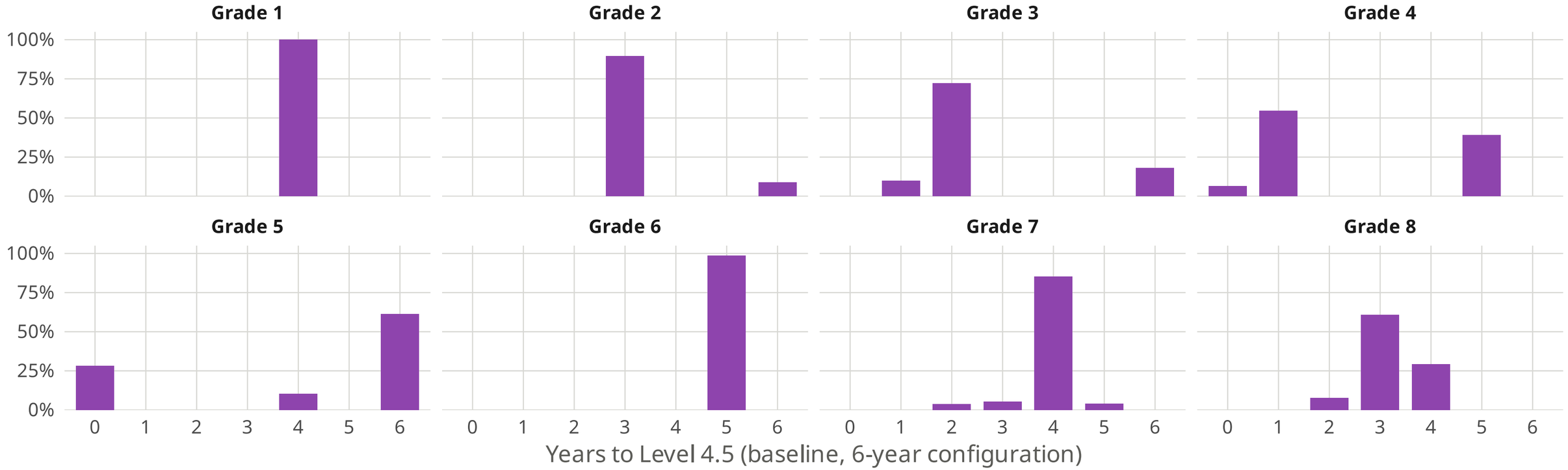
Current level	Catch-up rate	N
L1	0.4%	471
L2	8.7%	835
L3	38.5%	1,719
L4	69.4%	756
L4.5	100.0%	383
L5	100.0%	212
L6	100.0%	28

Baseline 6-year catch-up, old scale, 2026. The 6-year CUKU column codes students who started the window below 4.5; students already at the exit standard do not receive a 6-year Keep Up status. Cells with N < 10 are suppressed.

Years to the Level 4.5 target

Distribution of years-to-target, grades 1–8

From CATCH_UP_KEEP_UP_STATUS_BASELINE_6_YEAR / 6-year configuration. Zero = already at or above the exit standard.



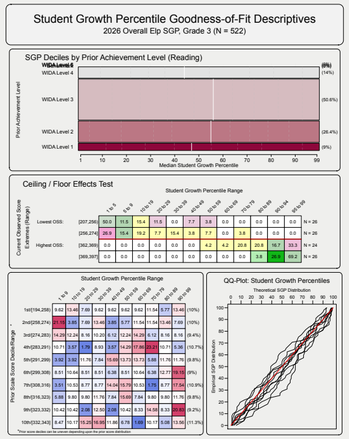
Cohort vs. baseline on-track — a discussion item

Do not treat these two columns as a trend

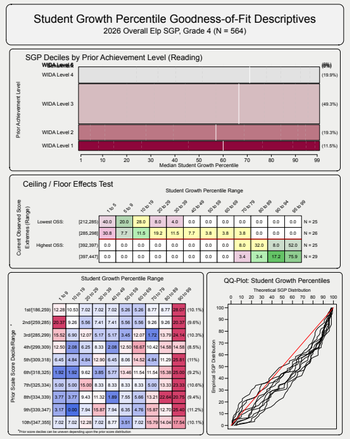
Baseline catch-up projects on the old scale and compares with the current 4.5 cuts. Cohort catch-up feeds each 2026 new-scale score into 2026 matrices whose prior-score slots were fit on old-scale scores, then compares the projections with the same old-scale cuts. The difference is a scale-transition artefact, not a finding.

Grade	Baseline catch-up (old scale)	Cohort catch-up (new scale)
1	28.0% (N = 443)	84.2% (N = 443)
2	73.2% (N = 511)	87.1% (N = 511)
3	57.7% (N = 522)	78.7% (N = 522)
4	62.4% (N = 564)	77.1% (N = 564)
5	40.7% (N = 337)	77.4% (N = 337)
6	43.7% (N = 286)	77.6% (N = 286)
7	46.2% (N = 318)	81.8% (N = 318)
8	41.2% (N = 320)	85.3% (N = 320)
9	33.6% (N = 336)	83.9% (N = 336)
10	13.3% (N = 286)	70.3% (N = 286)
11	12.5% (N = 256)	62.1% (N = 256)

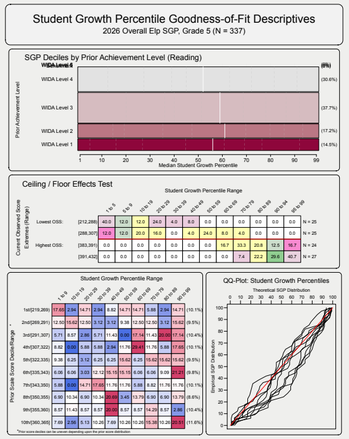
WIDA ACCESS – baseline model fit, 2026



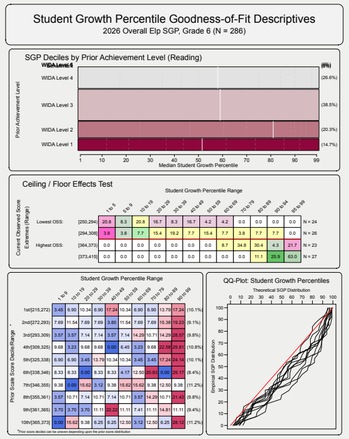
Grade 3



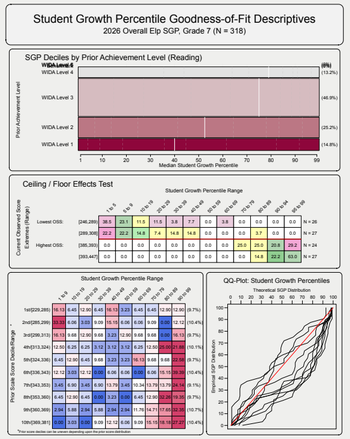
Grade 4



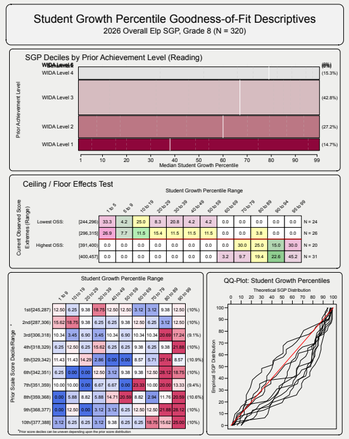
Grade 5



Grade 6



Grade 7



Grade 8

Summary

- **NH SAS cohort SGPs** show even conditional coverage across grades 4–8 in ELA and mathematics — suitable for accountability calculations.
- **NH SAS baseline SGPs** are the diagnostic lens on recovery relative to 2018–19. Grades flagged more than two points from 50 are listed on the baseline-results slide.
- **Within-year growth is close to comparable across student groups**; where it is not, that pair is the priority.
- **Catch-up / keep-up rates are not comparable across grades** without their target horizon, which shrinks toward grade 8.
- **WIDA ACCESS exit remains WIDA Level 4.5.** Headline on-track results are **baseline-referenced and computed on the old composite scale**.
- **2026 is an ACCESS 3.0 transition year.** Cohort targets on the new scale against old cuts should not be reported as a trend. Whether `WIDA_NH` adopts the consortium's 2026 transition metadata is a TAC question.

Questions for the TAC

Where the committee's guidance is needed

1. **How long should New Hampshire carry the old-scale WIDA baseline** after ACCESS 3.0? The 2026 split run is possible only because both composites are on the file.
2. **Should `WIDA_NH` adopt the consortium 2026 transition metadata now** (cuts, knots, linkages already on the generic `WIDA` entry), or wait until new standards are set? If those standards smooth the grade-cluster seams, grade 3–5 exit rates will move.
3. **Should 2026 cohort on-track results be withheld** from public reporting, the way a contradicted baseline series would be, until the new-scale cuts exist?
4. **Should the 2018–19 NH SAS baseline continue** as the monitoring series, or is it time to discuss a dual track — a fixed recovery series plus a re-established reporting norm?

Thank you

NEW HAMPSHIRE TECHNICAL ADVISORY COMMITTEE

New Hampshire 2026 SGP Results

Damian W. Betebenner · National Center for the Improvement of Educational Assessment